

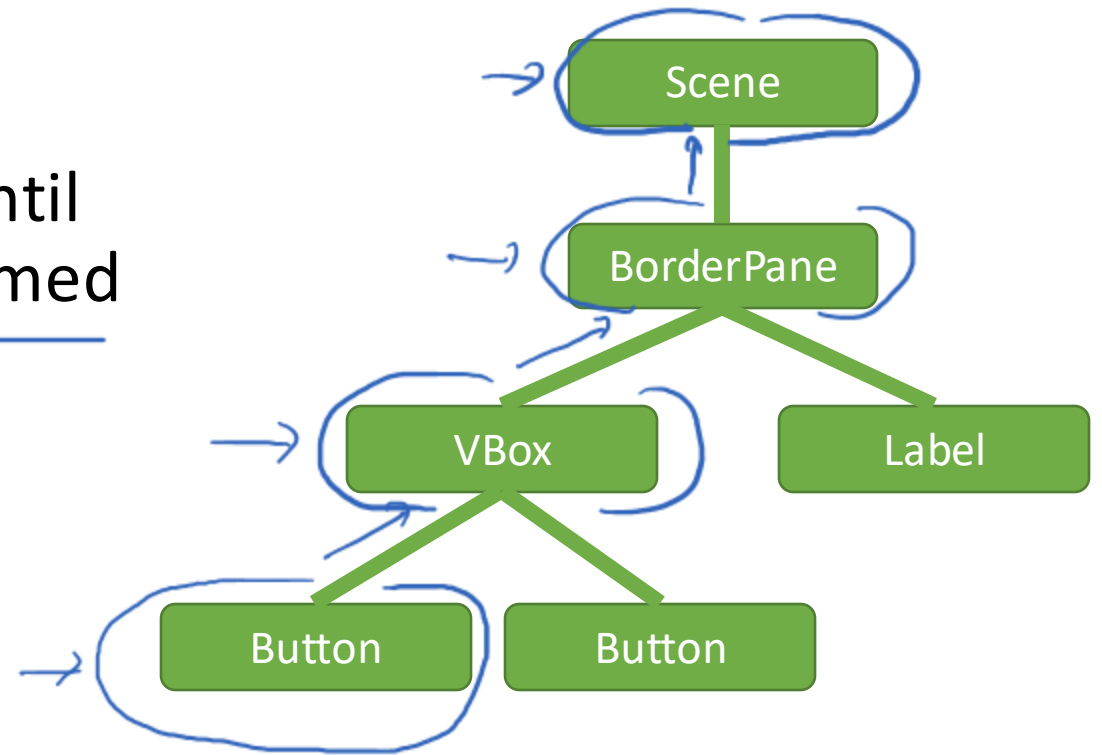
JavaFX Event Handling

What is an Event?

- A Java object that represents a user interaction with the GUI and contains data about the interaction
- A subtype of *javafx.event.Event*
(examples: MouseEvent, KeyEvent, ActionEvent)

Events in the Scene Graph

- Event starts at the control that the user interacts with (the target)
- It then moves up the scene graph until it reaches the root node or is consumed



Event Handlers

- Objects of type *javafx.event.EventHandler*
- Can be registered with a nodes in the scene graph
- Contain a method that gets called to process an event passing through the node they are registered with

The *EventHandler* Interface

```
public interface EventHandler<T extends Event> {  
  
    public void handle (T event);  
  
}
```

Registering EventHandlers

- Scene graph nodes have method

.addEventHandler(EventType<T> eventType,
EventHandler<? super T> eventHandler)

- Examples:
 - .addEventHandler(KeyEvent.KEY_TYPED, EventHandler<KeyEvent> handler)
 - .addEventHandler(MouseEvent.MOUSE_CLICKED, EventHandler<MouseEvent> handler)
 - .addEventHandler(ActionEvent.ACTION, EventHandler<ActionEvent> handler)